

A Strain-Resolved Time-Course Transcriptomic Analysis Identifies a Late H7N9-Specific Program Linking Metabolic Collapse to Reduced MHC II Activity

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- *code is shared for each person

Abstract

Avian influenza A viruses are a persistent zoonotic threat due to their zoonotic potential and capacity to cause severe disease. Since 2013, H7N9 has been linked to severe human pneumonia and high mortality. Evidence suggests severity reflects not only viral replication but also dysregulated host immune regulation and immunopathology. However, the timing of these alterations and what is truly H7N9-specific across strains remains unclear.

We analyzed a time-resolved Calu-3 transcriptomic dataset infected with H7N9, H3N2, H7N7, or VN1203 at 3, 7, 12, and 24 h. Time-matched mock controls were used to define infection-induced differential expression at each time point. Global structure was assessed by PCA/MDS, and strain-specific 24 h DEGs were defined by excluding genes shared with other viruses. GO/KEGG/Reactome enrichment and module scoring (metabolism, ER/Glyco/Lys, MHC I/II, ISR) supported functional interpretation.

Global ordinations showed virus- and time-dependent trajectories, with increasing divergence from mock and a large shared DEG core at 24 h. Despite this shared response, each strain retained distinct functional programs, and H7N9 displayed a unique 24 h signature. H7N9 specifically upregulated transcription/cell-cycle and lysosome-related pathways while downregulating central metabolic processes and innate effector terms. Within H7N9, antigen-presentation pathways (especially MHC II) were suppressed early despite interferon activation, followed by late metabolic collapse and ISR induction.

H7N9 follows a distinct temporal logic in airway epithelium, marked by early uncoupling of antiviral signaling from antigen presentation. At 24 h, H7N9 uniquely couples metabolic/ER-Glyco-Lys downregulation with reduced MHC II module activity. This profile contrasts with strains such as H3N2 that show stronger antigen-presentation engagement at late stages. Time-resolved comparative transcriptomics thus prioritizes late H7N9-specific pathways as candidates for mechanisms and therapeutic targeting.

Introduction

Avian influenza A viruses (IAVs) constitute a major reservoir of genetic diversity and remain a persistent threat to global health through zoonotic spillover and subsequent host adaptation. Since its emergence in 2013, avian-origin A(H7N9) has been associated with severe human lower respiratory tract disease and substantial mortality among hospitalized cohorts, despite early epidemic strains being characterized as low-pathogenic in poultry, underscoring a dissociation between avian pathogenicity and human disease severity.^[1,2] This observation has motivated intensive investigation into host determinants of disease, as accumulating clinical and experimental evidence suggests that adverse outcomes in H7N9 infection reflect not only viral replication but also qualitative differences in host immune regulation and immunopathology.^[2–4]

From an immunological standpoint, a recurring theme is an imbalanced innate response in which antiviral interferon (IFN) programs and inflammatory cytokine/chemokine outputs are not proportionally coordinated. Patient studies have reported elevated pro-inflammatory mediators—particularly IL-6 and chemokines such as CXCL10 (IP-10)—with levels associated with disease severity, supporting a clinically relevant hyperinflammatory component.^[2,5] In parallel, mechanistic and systems-level studies in controlled models have indicated that H7N9 can elicit host-response programs that differ from both seasonal IAVs and other avian strains, including strain-specific transcriptional signatures and distinct pathway engagement.^[3,6] Importantly, innate immune dysfunction in H7N9 is not limited to soluble mediators: inflammasome-associated pathways have been implicated as contributors to lung immunopathology. In a murine model, H7N9 infection activated the NLRP3–caspase-1 axis, and disruption of inflammasome/IL-1 signaling attenuated disease severity and improved survival, supporting a pathogenic role for excessive inflammasome signaling under certain conditions.^[7]

Despite these advances, the temporal structure of immune dysregulation—and its strain specificity—remains incompletely resolved. Many studies focus on single time points or single strains, limiting the ability to distinguish (i) shared antiviral programs that represent conserved host defense from (ii) late-emerging, strain-specific immune pathways that may track with pathogenic potential. Time-resolved, cross-strain transcriptomic profiling in relevant airway epithelial models provides a principled framework to address this gap. Notably, a comparative Calu-3 infection system has been used to profile host transcriptional responses to H7N9 alongside H3N2, H7N7, and VN1203 across 3, 7, 12, and 24h post-infection, enabling direct evaluation of early antiviral modules versus late inflammatory and innate effector pathways under matched experimental conditions.^[6] We leveraged a time-resolved Calu-3 transcriptomic dataset (3–24 h) comparing H7N9 with H3N2, H7N7, and VN1203 to define the global host-response landscape and pinpoint virus-specific pathway differences, with a focus on 24 h post-infection. We then reconstructed the H7N9 trajectory over time, revealing a late-stage program marked by a metabolic shift coupled to reduced antigen-presentation module activity.

Materials and Methods

Transcriptomic Dataset and Experimental Design

Gene expression data were obtained from human bronchial epithelial Calu-3 cells infected with four influenza A viruses—H3N2, H7N7, H7N9 (Anhui/01), and VN1203 (H5N1)—or mock infected. Infections were sampled at 03 h, 07 h, 12 h, and 24 h post-infection, yielding a total of 40 samples. All analyses were conducted in R (v4.x).

Data Preprocessing, Normalization, and Quality Control

Raw gene expression values were converted to numeric format, and gene identifiers were made unique. Expression values were $\log_2$ -transformed using $\log_2(x + 1)$ to stabilize variance and improve comparability across samples. Global expression distributions were assessed using boxplots to verify consistency across samples and conditions. No samples were excluded based on quality-control metrics.

Differential Expression Analysis

Differential expression analysis was primarily performed using the limma framework, which applies linear modeling and empirical Bayes moderation to improve variance estimation across genes. For virus-versus-mock comparisons, $\log_2$ -transformed expression values were modeled with virus infection as the main factor, and contrasts were defined for each virus relative to time-matched mock controls at 03 h, 07 h, 12 h, and 24 h. For selected analyses requiring direct pairwise comparisons (e.g., Venn diagram construction and virus-to-virus contrasts), gene-wise two-sided Student's *t*-tests were performed in $\log_2(x + 1)$ space. $\log_2$ fold change ($\log_2FC$) was calculated using the standard definition:

$$\log_2FC = \log_2(X_{\text{virus}} + 1) - \log_2(X_{\text{mock}} + 1)$$

Multiple-testing correction was applied using the Benjamini–Hochberg procedure. Genes were defined as differentially expressed if they met both an adjusted *P* value (FDR) < 0.05 and $|\log_2FC| > 1$.

Sample–Sample Correlation Analysis

To assess reproducibility among biological replicates and global relationships between conditions, Pearson correlation coefficients were computed using $\log_2$ -transformed expression values. Correlation matrices were visualized using the pheatmap package, with hierarchical clustering applied using Euclidean distance. Sample annotations included virus subtype and time point.

Principal Component Analysis (PCA)

PCA was performed to obtain an unsupervised overview of global transcriptional variation. The expression matrix was structured with samples as observations and genes as variables, and values were scaled so that each gene contributed equally. Virus subtype and time point were used solely for visualization and interpretation. PCA scores were plotted along the first two principal components, with samples colored by virus and labeled by time point. Ninety-five percent confidence ellipses were overlaid for each virus group to summarize within-group dispersion. Multidimensional scaling (MDS) was additionally performed using Euclidean distances computed from $\log_2$ -transformed expression values to provide an independent ordination of global transcriptional relationships across samples.

Venn Diagram Analysis of Differentially Expressed Genes

To examine overlap among transcriptional responses, DEGs were identified for each virus relative to time-matched mock controls at each time point. DEG lists were compiled independently for H3N2, H7N7, H7N9, and VN1203, and overlaps were visualized using four-set Venn diagrams generated with ggVennDiagram. Region counts were displayed, and figures were exported at 300 dpi. For H7N9-focused analyses, additional Venn diagrams were generated to examine temporal overlap of DEGs across 03 h, 07 h, 12 h, and 24 h.

Pathway Enrichment Analysis

Gene Ontology (GO) Biological Process enrichment was performed separately for upregulated and downregulated gene sets using clusterProfiler (enrichGO) with the org.Hs.eg.db annotation package (keyType = "SYMBOL"). Benjamini–Hochberg correction was applied, and terms with adjusted $P < 0.05$ were considered significant. Enrichment results were visualized as faceted bar plots showing $-\log_{10}(\text{adjusted } P \text{ value})$ and gene counts across time points. KEGG pathway enrichment analysis was performed in parallel using enrichKEGG, following conversion of gene symbols to Entrez identifiers, and is presented in Supplementary Figure S4. Reactome pathway enrichment analysis was performed in parallel using the ReactomePA package, with Benjamini–Hochberg–adjusted P values < 0.05 used to define significantly enriched pathways.

Progression-Based Differential Expression and Volcano Analysis

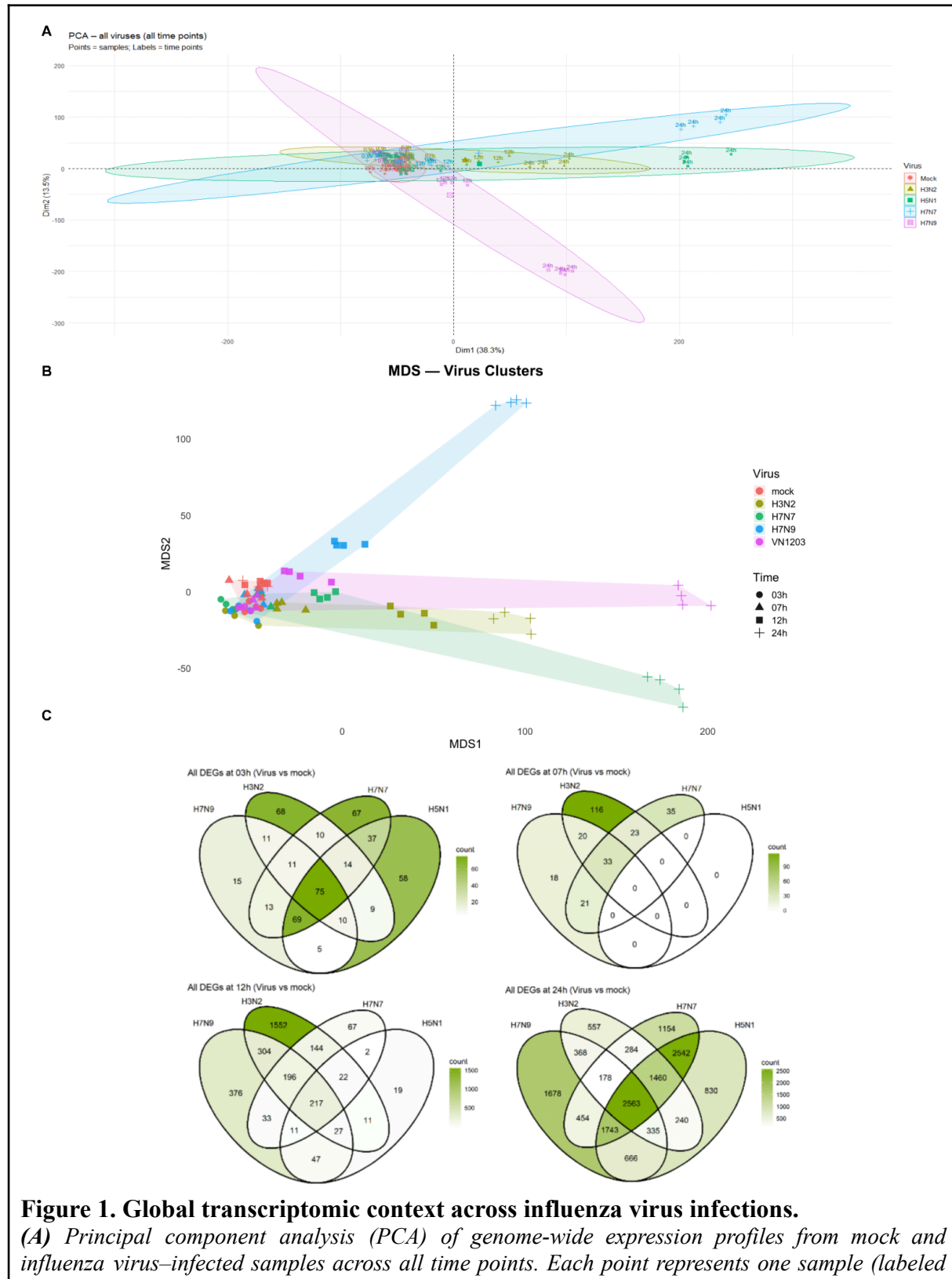
To quantify transcriptional progression during H7N9 infection, a within-virus contrast comparing H7N9 at 24 h versus 03 h was applied using the same differential expression framework. Genes meeting $\text{FDR} < 0.05$ and $|\log_2\text{FC}| > 1$ were classified as progressively upregulated or downregulated and visualized using volcano plots showing $\log_2\text{FC}$ versus $-\log_{10}(\text{FDR})$. Selected interferon-stimulated, inflammatory, and metabolic genes were annotated to illustrate immune activation and metabolic suppression.

Module Scoring and Coupling Analysis

Module activity was quantified from the $\log_2$ -transformed expression matrix using five predefined gene sets: Metabolism, ER/Glyco/Lys, MHC-I, MHC-II, and Integrated Stress Response (ISR). Only genes present in the dataset were retained. Expression values were Z-scored gene-wise across all samples, and module scores were calculated as the mean Z-score of genes within each module for each sample. Module scores were merged with sample metadata (virus and time point). Time-course analyses were performed by plotting mean module scores per virus at each time point. To quantify infection-induced changes, Δ versus mock values were calculated by subtracting the mean mock module score from each infected sample at the corresponding time point. Heatmaps and bar plots were used to visualize strain-specific module activity at 24 h post-infection.

Results

I. Global transcriptomic context



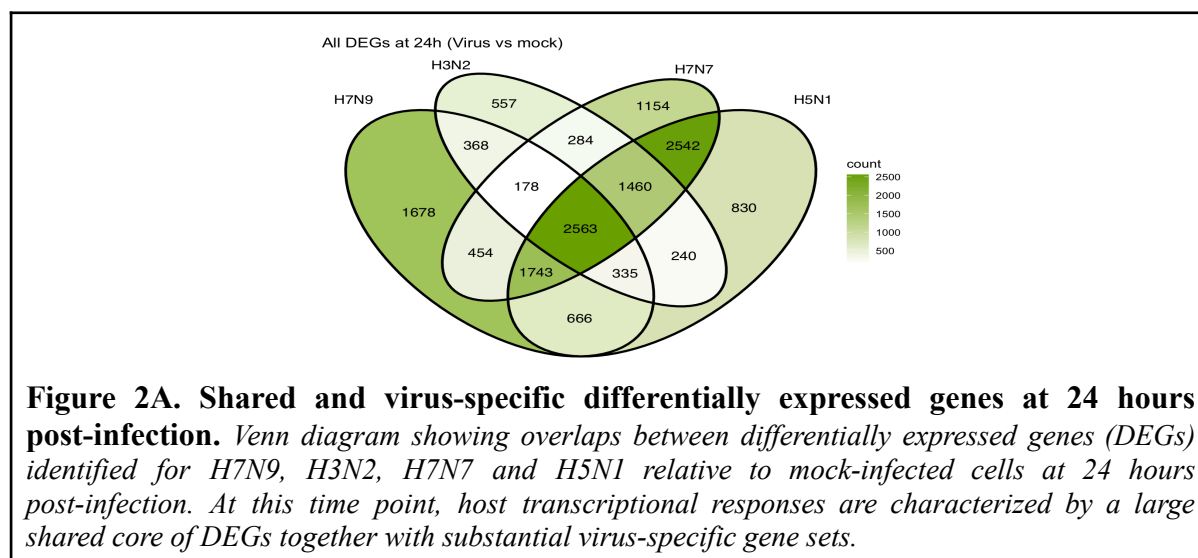
by time point), and ellipses indicate the dispersion of each condition in PCA space. The first two components (Dim1 = 38.3%, Dim2 = 13.5%) capture major sources of variation driven by both virus identity and infection progression: mock samples remain tightly clustered near the origin, whereas infected samples separate into virus-specific trajectories, with later time points showing greater displacement from the center. **(B)** Multidimensional scaling (MDS) based on global expression distances provides an independent ordination that recapitulates the virus-structured clustering and time-dependent divergence observed by PCA, supporting the robustness of these global relationships. **(C)** Venn diagrams summarizing differentially expressed genes (DEGs; virus vs time-matched mock) at 03 h, 07 h, 12 h, and 24 h. Numbers denote DEG counts in each overlap region (color intensity reflects magnitude). Across viruses, DEG burden and cross-virus overlap increase over time, with a small early shared response and a substantially larger shared core by 24 h, alongside pronounced virus-specific components.

To place downstream analyses in a global context, we first examined genome-wide transcriptional variation across mock and infected samples over time. PCA revealed that global expression patterns are jointly structured by virus identity and infection progression (Fig. 1A). Mock samples clustered tightly near the origin, whereas infected samples followed virus-specific temporal trajectories that diverged progressively from the central cluster as infection advanced. Notably, H7N7 exhibited the strongest global displacement predominantly along Dim1, with 24 h samples occupying the most extreme positions on this axis, consistent with a large-amplitude transcriptomic reprogramming. In contrast, H7N9 separated along a distinct axis, showing prominent movement along Dim2 (negative direction) at later time points, indicating a qualitatively different global response relative to the other strains. H3N2 remained comparatively closer to the origin across the time course, consistent with a more moderate global perturbation, while H5N1 displayed a clear but intermediate directional shift that became more pronounced at later time points. An independent ordination using MDS reproduced the same virus-structured clustering and time-dependent divergence, supporting the robustness of these global relationships (Fig. 1B).

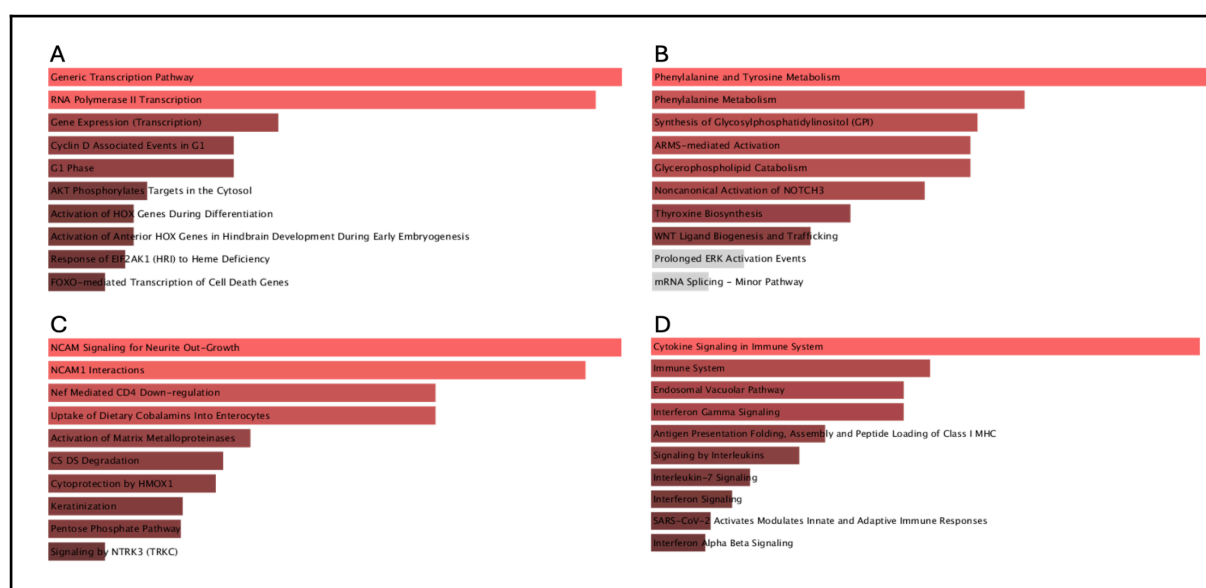
We next quantified how differential expression evolved across viruses over time using virus vs time-matched mock contrasts and summarized DEG overlap patterns (Fig. 1C). At 03 h, all four viruses shared a small common response, alongside modest virus-specific components, consistent with an early response composed of a limited shared core plus strain-specific transcriptional features. By 07 h, the response remained relatively constrained and strongly strain-dependent: H3N2 showed the largest virus-specific component, H7N9 and H7N7 retained smaller unique sets, while H5N1 contributed little to no virus-specific signal at this time point, suggesting delayed or more subtle early transcriptional changes relative to the others. Notably, H5N1 (VN1203) yielded few to no significant DEGs at 07 h under our gene-wise *t*-test framework; given the small number of biological replicates and the absence of empirical Bayes variance moderation, this most likely reflects limited statistical power after multiple-testing correction rather than a complete lack of transcriptional perturbation at this stage. At 12 h, the transcriptional response expanded, with H3N2 showing a pronounced virus-specific component, whereas H7N9 increased more moderately, and H7N7 and H5N1 remained comparatively limited; the shared four-way component also increased, indicating an emerging common host program. By 24 h, all viruses showed extensive differential expression with simultaneous convergence and diversification: the shared four-way overlap increased dramatically to 2,563 DEGs, consistent with a large common late host-response module, while substantial virus-specific components persisted and differed by strain, indicating that late infection is characterized by both a strong conserved transcriptional signature and pronounced strain-specific remodeling. Collectively, the PCA/MDS ordinations together with time-resolved DEG overlap demonstrate that influenza viruses elicit distinct

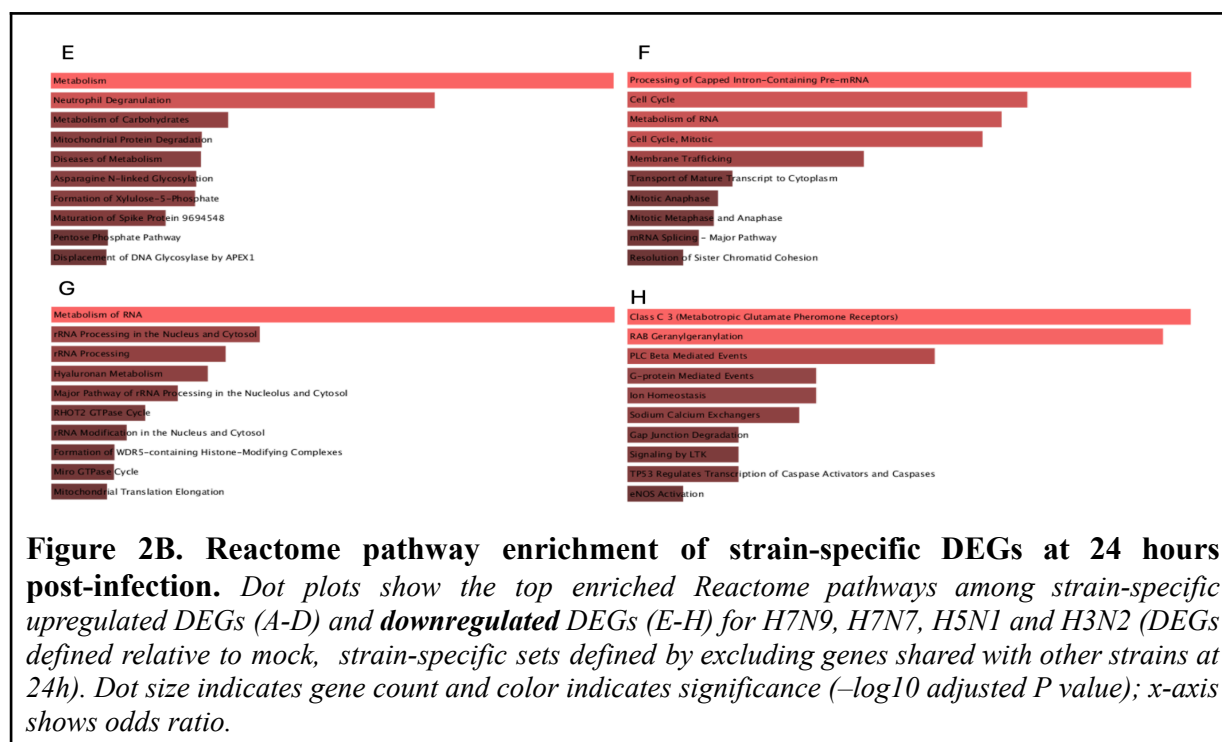
global transcriptional trajectories that diverge from mock in a time-dependent manner, while progressively accumulating a shared late response that coexists with persistent virus-specific programs (Fig. 1A–C).

II. Comparative pathway analysis at 24h post-infection reveals distinct virus-specific functional responses



At 24 hours post-infection, all influenza viruses induced extensive transcriptional remodeling and shared a large core of DEGs across strains (**Fig. 2A**). In addition, each virus retained substantial strain-specific DEG subsets, indicating divergence in strain-associated transcriptional programs at 24h post infection. Notably, H7N9 displayed a large set of DEGs not shared with the other viruses, together with subsets overlapping with one or two strains. To resolve the functional composition of these strain-specific transcriptional programs, we performed pathway enrichment analyses on strain-specific upregulated and downregulated gene sets for each virus (**Fig. 2B**). Consistent with these strain-specific profiles, virus-to-virus comparisons at 24 h (excluding mock) defined a robust H7N9-relative signature, with enrichment patterns again highlighting increased transcriptional programs and reduced immune-associated pathways relative to other strains (Supplementary Fig. II).





For H7N9, enrichment among strain specific upregulated genes was dominated by transcription-associated programs, including RNA polymerase/transcription-related pathways and generic transcriptional regulation, together with G1-phase cell-cycle modules and lysosome pathways. In contrast, H7N9 strain-specific downregulated genes were enriched for central metabolic pathways, including carbohydrate metabolism and mitochondrial-associated programs, as well as innate effector modules, including antimicrobial responses such as neutrophil degranulation.

H7N7 exhibited a distinct enrichment profile. Strain-specific upregulated genes were enriched for pathways related to metabolic homeostasis and intracellular trafficking, including amino acid and lipid metabolism, mitochondrial function, and vesicular transport. Strain-specific downregulated genes were enriched for pathways involved in cell cycle progression and RNA maturation.

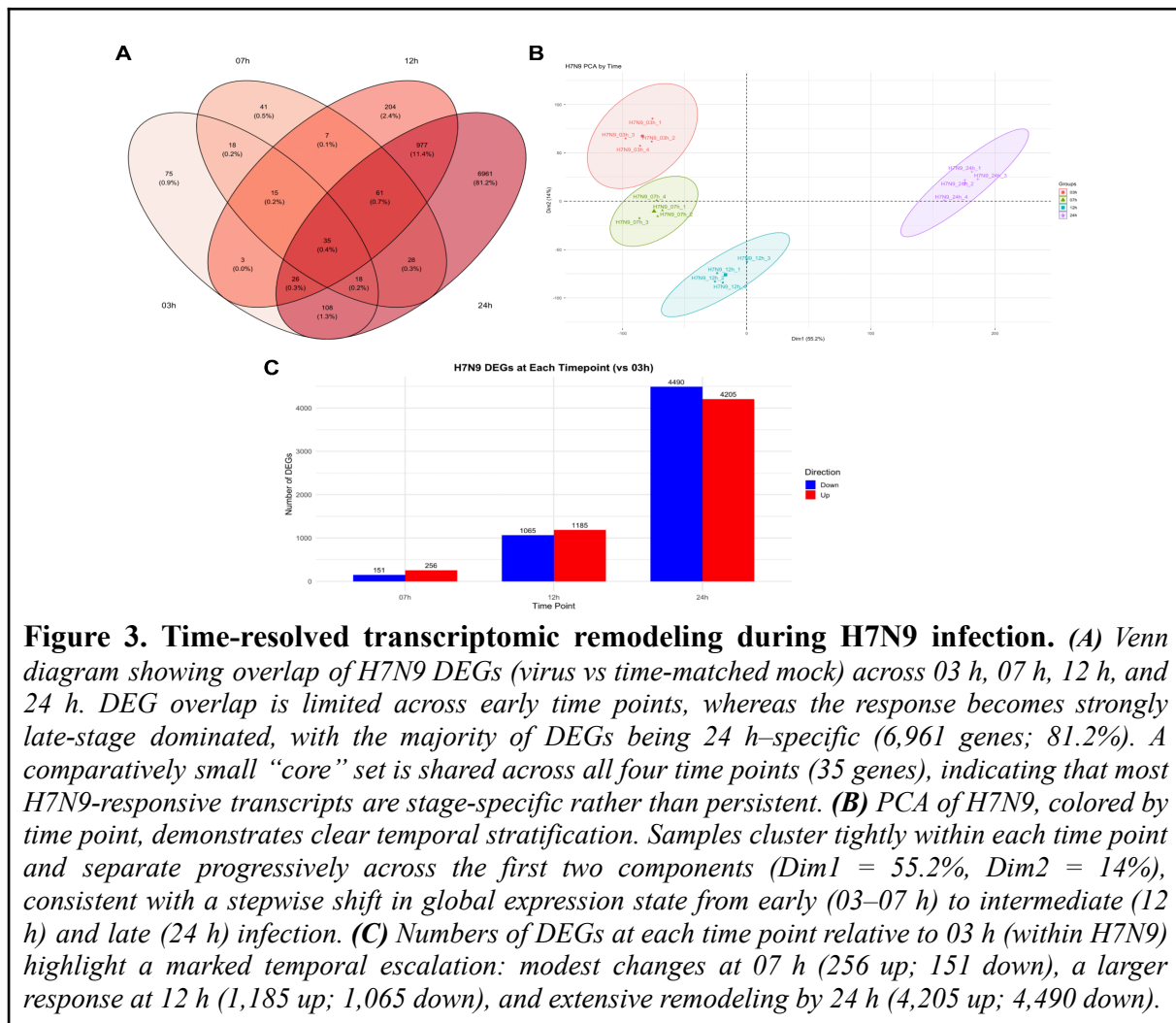
H5N1 strain-specific upregulated genes were enriched for pathways involved in tissue remodeling, stress related processes, and immune modulation, including extracellular matrix-associated programs. Conversely, H5N1 strain-specific downregulated genes were enriched for RNA metabolism and mitochondrial translation.

H3N2 displayed a contrasting enrichment pattern dominated by immune and antiviral programs. Strain-specific upregulated genes were enriched for interferon mediated signaling, cytokine responses, and antigen processing and presentation. Strain-specific downregulated pathways were comparatively limited and included signaling and communication modules such as gap junction pathways and class C/3 (metabotropic glutamate/pheromone receptor-related) signaling.

A direct comparison between H7N9 and H3N2 strain-specific enrichment profiles highlighted distinct pathway distributions: H3N2 strain-specific upregulated signatures were concentrated in interferon signaling and antigen presentation modules, whereas H7N9 strain-specific upregulated signatures were concentrated in transcription/cell-cycle and lysosome-associated programs. Conversely, H7N9 strain-specific downregulated signatures were dominated by metabolic and innate effector pathways, while H3N2 strain-specific downregulated signatures were fewer and mapped primarily to signaling and communication pathways.

Together, these analyses show that, despite a shared transcriptional core at 24 hours post-infection, each influenza strain is associated with a distinct strain-specific functional profile defined by pathway enrichment among strain-specific upregulated and downregulated genes.

III. Time analysis of H7N9



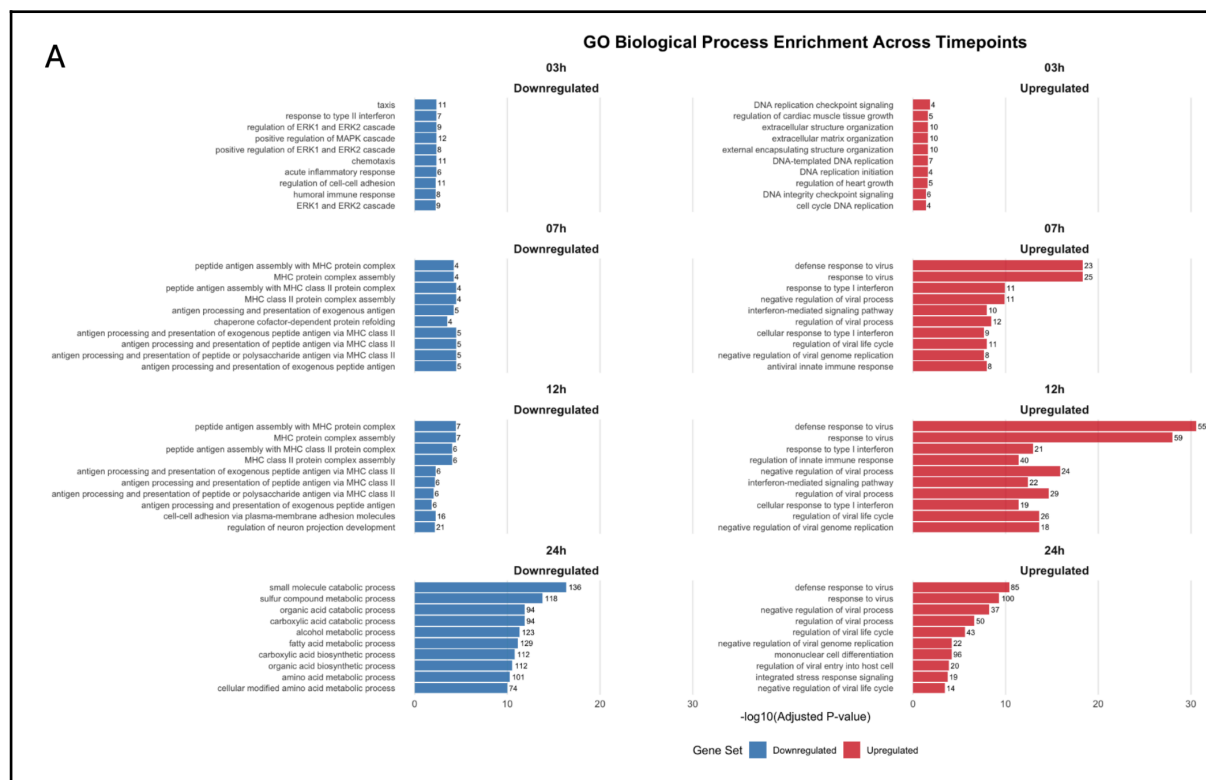
Given the distinctive global trajectory of H7N9 observed in the multi-virus ordinations, we next performed a focused time-resolved analysis within H7N9 to determine whether its response reflects a stable core program or a sequence of stage-specific transcriptional states. PCA restricted to H7N9 samples revealed a clear and reproducible temporal stratification, with samples clustering tightly by time point and separating progressively across the first two components (Dim1 = 55.2%, Dim2 = 14%; Fig. 3B). The spatial ordering of the clusters

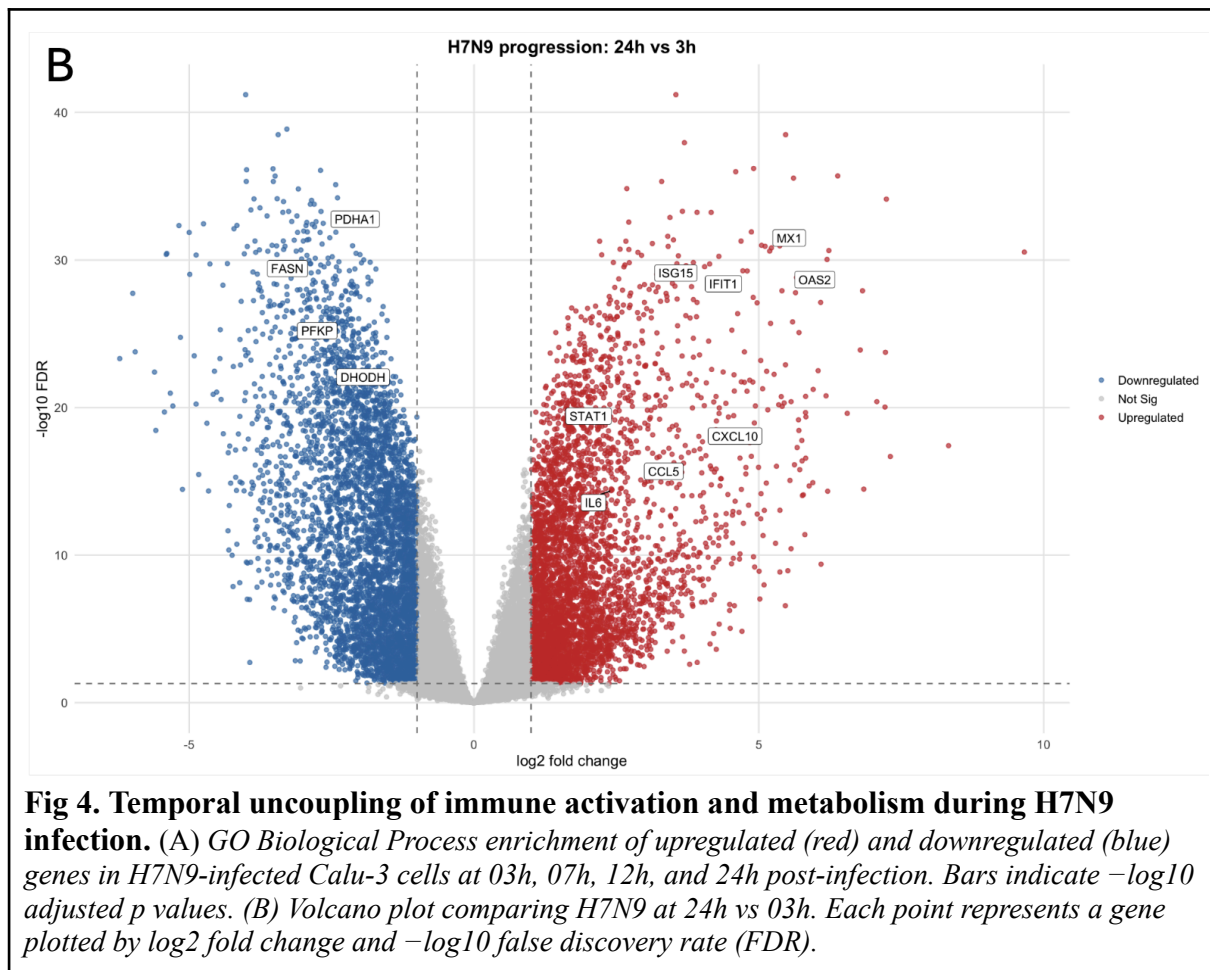
indicates a stepwise transition in the global expression state: early samples (03–07 h) remain relatively proximal, 12 h samples occupy a distinct intermediate position, and 24 h samples are most distant, consistent with substantial late-stage transcriptomic divergence.

To quantify the stage-specificity of H7N9 responses, we compared H7N9-infected samples to time-matched mock controls at each time point and examined overlap among DEG sets (Fig. 3A). This analysis demonstrated that the transcriptional response is dominated by late infection, with 6,961 genes (81.2%) uniquely detected at 24 h, whereas time point–exclusive DEG sets at earlier stages were comparatively small (03 h: 75; 07 h: 41; 12 h: 204). Overlap across multiple time points was limited, and only 35 genes were shared across all four stages, indicating that most H7N9-responsive genes are temporally restricted rather than uniformly maintained throughout infection. Notably, the largest shared component occurred between 12 h and 24 h (977 genes), suggesting that a substantial fraction of the late response emerges during the 12→24 h transition, rather than being fully established early.

In parallel, to directly measure how far the H7N9 transcriptome moves away from an early reference state, we performed within-virus contrasts using 03 h as baseline (Fig. 3C). The number of DEGs increased sharply over time, rising from a modest response at 07 h (256 upregulated; 151 downregulated) to a broader response at 12 h (1,185 up; 1,065 down), and culminating in extensive remodeling by 24 h (4,205 up; 4,490 down). The comparable scale of induction and repression at 24 h indicates that late infection involves coordinated activation and shutdown of large gene programs rather than a unidirectional shift. Together, these complementary analyses (virus vs mock overlap and 03 h-referenced temporal contrasts), supported by the H7N9-only PCA, demonstrate that H7N9 triggers a strongly time-dependent host response characterized by limited early perturbation, a transitional mid-stage, and broad late-stage reprogramming dominated by largely 24 h–specific differential expression (Fig. 3A–C).

IV. H7N9 causes late metabolic shift





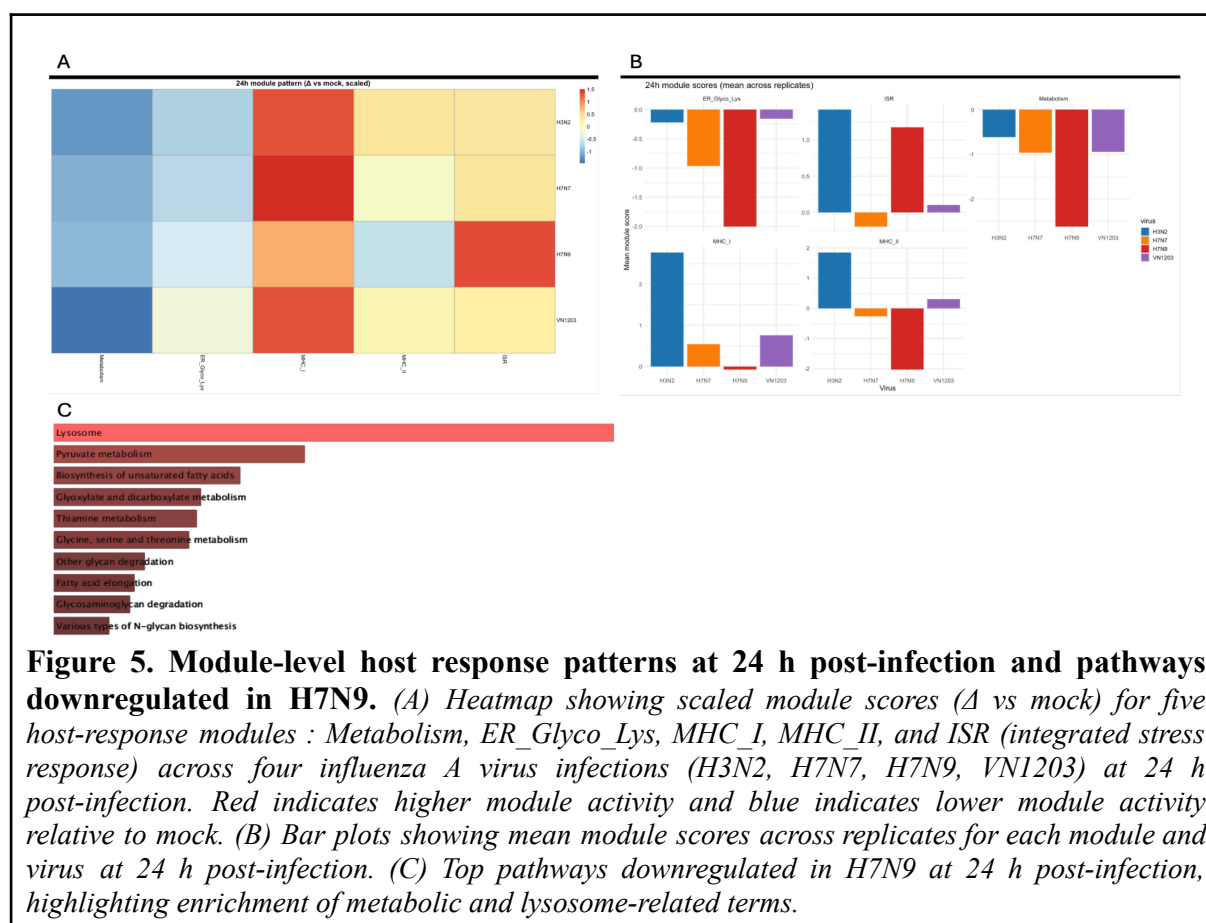
To define the temporal coordination between immune activation, antigen presentation, and host metabolism during H7N9 infection, we performed time-resolved Gene Ontology (GO) Biological Process enrichment analysis together with a progression-based differential expression analysis. Across early and intermediate timepoints (03h–12h post-infection), H7N9 infection was characterized by a consistent and directional suppression of antigen presentation pathways. GO enrichment of downregulated genes repeatedly identified processes related to MHC class II protein complex assembly, antigen processing and presentation via MHC class II, and peptide antigen assembly with MHC protein complexes at 03h, 07h, and 12h, indicating that antigen presentation machinery is transcriptionally suppressed from the earliest stages of infection and remains repressed through mid-infection. In parallel, enrichment analysis of upregulated genes at 07h and 12h revealed strong activation of innate antiviral programs, dominated by defense response to virus, type I interferon signaling, and interferon-mediated antiviral pathways. KEGG pathway enrichment independently confirmed this pattern, with antigen processing and presentation pathways significantly downregulated at mid timepoints, while viral and interferon-associated pathways were upregulated (Supplementary Figure S4).

By 24h post-infection, the transcriptional response shifted markedly. Downregulated genes became overwhelmingly enriched for metabolic processes, including small molecule catabolism, fatty acid metabolism, amino acid metabolism, organic and carboxylic acid metabolism, and glycoprotein metabolic processes, consistent with a broad collapse of cellular

metabolic capacity. Concurrently, upregulated genes at 24h were enriched for pathways associated with cellular stress and viral persistence, including integrated stress response signaling. KEGG analysis corroborated these findings, revealing suppression of core metabolic pathways such as carbon metabolism, glycolysis/gluconeogenesis, fatty acid metabolism, and lysosomal function (Supplementary Figure S4), indicating that metabolic and lysosomal dysfunction dominate the late transcriptional landscape.

To directly capture the progression of these responses, we compared H7N9 at 24h versus 03h. The resulting volcano plot revealed a clear dichotomy in gene regulation, with progressively upregulated genes dominated by interferon-stimulated and inflammatory mediators (including STAT1, ISG15, MX1, OAS2, CXCL10, CCL5, and IL6), while progressively downregulated genes were enriched for metabolic enzymes central to energy production and biosynthesis (including PFKF, FASN, PDHA1, and DHODH). Together, these analyses define a coherent temporal trajectory in which H7N9 infection induces early and sustained suppression of antigen presentation alongside robust antiviral signaling, followed by late-stage metabolic collapse and integrated stress response, revealing a progressive uncoupling of immune activation from metabolic and antigen-presentation homeostasis.

V. H7N9 couples late metabolic downregulation with reduced antigen-presentation module activity



At 24 h post-infection, module-level changes revealed strain-specific patterns across Metabolism, ER_Glyco_Lys, MHC_I, MHC_II, and ISR (integrated stress response)(Fig. 5A). H7N9 showed the strongest coordinated decrease in Metabolism and ER_Glyco_Lys,

accompanied by a pronounced reduction in MHC_II and limited induction of MHC_I. In contrast, H3N2 displayed robust activation of antigen presentation modules, with strong positive shifts in both MHC_I and MHC_II. H7N7 and VN1203 exhibited intermediate profiles, with more modest perturbations across these modules compared with H7N9. Bar plots of mean module scores across replicates corroborated these trends (Fig. 5B) . H7N9 had the most negative scores for Metabolism and ER_Glyco_Lys, together with a marked decrease in MHC_II, whereas H3N2 showed the highest positive scores for MHC_I and MHC_II. The ISR module was strongly induced in H7N9 (and was also elevated in H3N2), while remaining comparatively low in H7N7 and VN1203. Focusing on H7N9 at 24 h, downregulated pathways were dominated by metabolic programs and included lysosome-related terms (Fig. 5C), consistent with the strong decrease in the Metabolism and ER_Glyco_Lys modules observed in panels A–B .

Together, these analyses indicate that H7N9 uniquely couples late metabolic downregulation with reduced antigen-presentation module activity, coinciding with decreased ER/Glyco/Lys module scores and a strong ISR signature.

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Discussions

Discussion - Shubhdeep Kaur

Human infection with avian-origin influenza A(H7N9) virus is associated with severe viral pneumonia, acute respiratory distress syndrome, and high mortality, despite early epidemic strains being classified as low pathogenic in poultry (1,2). This paradox has prompted investigation into host-driven determinants of disease severity. Clinical and virological studies established that viral loads alone do not fully account for the clinical course of H7N9 infection, with patient studies reporting elevated pro-inflammatory mediators and excessive inflammasome signaling associated with disease severity (3,4). However, the temporal structure of immune dysregulation and its strain specificity has remained incompletely resolved. To address this gap, we performed time-resolved transcriptomic profiling of human bronchial epithelial Calu-3 cells infected with Anhui01 (H7N9), NL219 (H7N7), Pan99 (H3N2), or VN1203 (H5N1) at different timepoints post-infection. This comparative framework identifies a distinct late-stage host state unique to H7N9 infection, characterized by coordinated metabolic shutdown, lysosomal and endoplasmic reticulum dysfunction, and impaired antigen-presentation capacity despite sustained antiviral signaling.

At early stages of infection, epithelial responses were broadly conserved across all influenza viruses examined. At 03h, global expression patterns were remarkably similar, with principal component and multidimensional scaling analyses showing that samples clustered primarily by time point rather than by viral strain. Differential expression analysis at this stage identified only a small number of genes altered across all four viruses, alongside modest virus-specific signatures, indicating a limited early transcriptional response dominated by shared host defense programs. By 07h, this pattern persisted, with modest strain-specific components beginning to emerge but overall transcriptional architecture remaining constrained. This conserved early response is consistent with prior work demonstrating that influenza viruses activate common innate sensing pathways in epithelial cells, including RIG-I-mediated recognition, NF- κ B activation, and induction of interferon-stimulated genes (5,6). Critically, H7N9 and seasonal H3N2 exhibited similar early transcriptional profiles despite their divergent pathogenic potential in humans, arguing against models in which H7N9 severity arises from exaggerated early interferon induction or fundamentally distinct initial immune evasion strategies.

Virus-specific divergence emerged progressively over time and became most pronounced at late stages of infection. While all viruses shared a substantial common transcriptional core at 24 hours post-infection—reflecting a conserved late host response to influenza—each strain retained distinct functional programs superimposed on this shared background. Seasonal H3N2 infection was characterized by enrichment of interferon signaling, cytokine-mediated communication, and antigen processing and presentation pathways, consistent with effective coordination between innate signaling and adaptive immune priming. In contrast, H7N9 infection exhibited a fundamentally different transcriptional architecture. Although antiviral and inflammatory genes were robustly induced, H7N9-specific downregulated genes were dominated by central metabolic pathways and lysosome-associated processes. This pattern was not observed to a comparable extent in H3N2, H7N7, or H5N1 infection, identifying metabolic and lysosomal suppression as defining features of the H7N9 host response.

Longitudinal analysis within H7N9 infection revealed that its transcriptional response was strongly late-dominated and stage-specific. Early and intermediate time points were characterized by limited transcriptional perturbation, whereas late infection involved

extensive remodeling encompassing both induction and repression of large gene programs. This balance between activation and shutdown indicates that late H7N9 infection is not simply an amplified immune response but rather a transition to a qualitatively distinct cellular state. These kinetics support a model in which H7N9 pathogenesis is driven by a progressive failure to restore cellular homeostasis, culminating in broad transcriptional collapse at later stages rather than by excessive early inflammation.

A particularly striking feature of the H7N9 response was the early and sustained suppression of antigen-presentation pathways, especially those associated with MHC class II. Antigen-presentation machinery was transcriptionally repressed from the earliest stages of infection and remained suppressed through mid-infection, even as interferon-mediated antiviral signaling was robustly induced. This dissociation reveals a fundamental uncoupling between innate immune activation and antigen-presentation capacity. Effective antiviral immunity requires tight coordination between epithelial sensing, cytokine production, and antigen presentation to support adaptive immune priming(7,8). Persistent suppression of antigen-presentation pathways may therefore impair T cell activation and delay immune resolution, consistent with clinical observations of impaired T cell responses and prolonged viral shedding in severe H7N9 infection (9).

By 24h post-infection, the H7N9 transcriptional response shifted markedly. Downregulated genes became overwhelmingly enriched for metabolic processes, including small molecule catabolism, fatty acid metabolism, amino acid metabolism, and glycoprotein metabolic processes. KEGG pathway analysis corroborated these findings, revealing suppression of carbon metabolism, glycolysis/gluconeogenesis, and lysosomal function. Concurrently, the integrated stress response (ISR) was strongly induced, as demonstrated by both pathway enrichment and module scoring analyses. The importance of cellular metabolism in immune function has become increasingly clear. Immune responses are energetically demanding, and metabolic reprogramming is required to sustain antiviral defense, antigen processing, and cytokine production (10,11). The metabolic shutdown observed in H7N9 infection suggests a failure to sustain these processes, likely constraining immune competence. Chronic ISR activation has been implicated in immune dysfunction and unresolved inflammation across multiple disease contexts (12). In H7N9 infection, ISR induction coincided with metabolic repression and lysosomal dysfunction, suggesting that epithelial cells enter a state of stress-induced functional paralysis rather than productive antiviral defense.

An additional layer of H7N9 specificity is supported by supplementary analyses that excluded mock controls and instead focused on genes commonly regulated across influenza viruses but quantitatively more extreme in H7N9. These analyses indicate that H7N9 pathogenicity is partly driven by a “setpoint shift,” in which shared antiviral and stress-response pathways are more strongly induced, while immune-associated and homeostatic programs are more deeply suppressed than in other strains. This quantitative imbalance reinforces a model in which severe H7N9 disease reflects dysregulated coordination of host responses rather than activation of entirely unique pathways.

Module-level analysis reinforced this interpretation by revealing coordinated suppression of metabolic, ER–glyco–lysosomal, and MHC class II modules uniquely in H7N9 infection at late stages. In contrast, H3N2 maintained or enhanced antigen-presentation module activity alongside relatively preserved metabolic function. Lysosomes and the endoplasmic reticulum occupy a central nexus linking metabolism and immunity, being essential for antigen

processing, peptide loading, and vesicular trafficking(13). Their coordinated suppression in H7N9 infection therefore provides a mechanistic bridge between metabolic failure and impaired antigen presentation, suggesting a tipping point beyond which effective immune resolution becomes unlikely.

Previous transcriptomic studies of H7N9 infection have reported exaggerated inflammatory signatures and identified host-response modules associated with pathogenic avian influenza viruses(3,6). However, many of these analyses focused on single time points or lacked direct cross-strain temporal comparisons, limiting the ability to distinguish conserved antiviral programs from late-emerging, strain-specific dysfunction. By explicitly integrating time-resolved and multi-strain comparisons within a unified framework, our analysis reveals that the defining features of the H7N9 response are not exaggerated early antiviral activation but rather a progressive failure to coordinate immune signaling with antigen-presentation and metabolic homeostasis. This temporal uncoupling is not apparent from static analyses and represents a key conceptual advance over prior work.

Several limitations should be considered when interpreting these findings. The analysis was performed in a single airway epithelial cell line and therefore does not capture the full multicellular and immune complexity of the lung in vivo. In addition, conclusions are based on transcriptomic inference without direct measurement of protein abundance, metabolic flux, or immune effector function, and the in vitro model lacks adaptive immune components required to assess T cell priming or viral clearance(7,8). These considerations underscore the need for complementary experimental systems to validate and extend the proposed mechanisms.

Future studies should directly test the functional consequences of immune–metabolic uncoupling during H7N9 infection using complementary models, including epithelial–immune co-cultures, airway organoids, and in vivo systems. Integrating proteomic and metabolomic analyses would validate the inferred metabolic and lysosomal dysfunction(10–13). Extending this temporal comparative framework to additional respiratory viruses may further clarify whether immune–metabolic collapse is a general feature of severe viral disease or a defining property of H7N9 pathogenesis.

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Discussion - Husein Kheil

Using a controlled, time-resolved infection series in human airway epithelium, our analysis separates a conserved influenza induced transcriptional core from strain specific programs that emerge on top of this shared response. Across H3N2, H7N7, VN1203 (H5N1) and H7N9, global ordinations (PCA/MDS) and DEG overlaps support a consistent structure: early time points (3–7 hpi) show limited and heterogeneous remodeling, whereas by 24 hpi all viruses share a large DEG core, consistent with canonical late antiviral and stress programs. Importantly, each strain still retains a distinct functional “overlay”, indicating that late infection combines shared host defense with virus specific reprogramming.

At 24 hpi, strain-specific DEG sets clarify what is genuinely “H7N9-specific”. While H3N2 specific signatures are enriched for interferon/cytokine signaling and antigen processing/presentation, H7N9 specific upregulated genes preferentially map to transcriptional regulation/cell-cycle programs and lysosome-related pathways, whereas H7N9 specific downregulated genes are dominated by central metabolism and immune-effector related terms. This argues that H7N9 does not simply induce “more” or “less” of a generic response, but instead shifts the balance toward transcriptional remodeling and endolysosomal biology while suppressing metabolic capacity and selected immune-associated outputs.

The prominence of lysosome-related enrichment within H7N9’s 24 hpi signature is particularly notable given accumulating evidence that endolysosomal routing and ubiquitin-dependent turnover can represent potent intrinsic antiviral control points. For example, MARCH8, an E3 ligase with established roles in membrane protein regulation ,

restricts influenza A replication by redirecting viral M2 from the plasma membrane to lysosomes, through K63-linked polyubiquitination at M2 K78. Viruses carrying K78R escape this restriction and show enhanced replication and pathogenicity in vivo¹.

Although our dataset does not directly test MARCH8–M2 dynamics, our enrichment patterns are consistent with the idea that influenza strain specific differences can be encoded at the level of endomembrane trafficking, lysosomal turnover, and host intrinsic restriction. In this framework, H7N9's lysosome-associated signature may reflect either (i) an amplified host attempt to engage degradative pathways during infection or (ii) viral remodeling of endolysosomal compartments with downstream consequences for secretion, antigen presentation, and cellular homeostasis. Distinguishing these alternatives will require targeted perturbations and protein-level tracking of endolysosomal cargo.

A second major conclusion arises from the H7N9 time course: H7N9 infection is best described as a sequence of stage specific transcriptional states rather than a stable “core program”. The Venn structure of H7N9 DEGs across 3–24 hpi indicates that most differentially expressed genes are detected primarily at 24 hpi, with limited persistence across all time points. Together with the H7N9-only PCA, these results support a model in which early sensing and antiviral signaling gradually transitions into a qualitatively different late host state. Notably, the most substantial overlap occurs between 12 and 24 hpi, suggesting that the critical reprogramming window is the 12 => 24 h transition, when sustained viral burden and cumulative stress may push infected epithelial cells into a distinct attractor characterized by resource limitation and stress-adaptation.

Within this temporal trajectory, our analyses identify an immunologically meaningful dissociation: H7N9 triggers robust interferon-associated antiviral programs while suppressing antigen-presentation pathways (particularly MHC II-associated terms) from early stages onward. This uncoupling is conceptually important because it suggests that epithelial innate activation is not automatically coupled to effective immune communication or adaptive priming signals. Although airway epithelial cells are not professional antigen-presenting cells, they contribute to immune coordination by expressing antigen presentation machinery and by shaping the chemokine/cytokine environment that recruits and instructs immune cells². An early, sustained reduction in antigen-presentation-related transcription could therefore bias the tissue response toward inflammation without proportional engagement of pathways that facilitate timely adaptive control. This interpretation aligns with human immunology studies of severe H7N9 showing that favorable outcomes correlate with early prominent H7N9-specific CD8 T cell responses, whereas fatal cases exhibit minimal influenza-specific immunity and little evidence of T cell activation³. While our Calu-3 transcriptome cannot capture lymphocyte kinetics directly, the epithelial state we observe is strong antiviral signaling paired with reduced antigen-presentation module activity.

At the late stage (24 hpi), H7N9 displays a striking convergence of module-level phenotypes: metabolism down, ER/Glyco/Lys down, ISR up, and MHC II module activity reduced, with comparatively limited induction of MHC I. This combined profile suggests that H7N9 infected epithelial cells enter a state of constrained bioenergetic and secretory capacity while engaging stress response pathways that reprogram translation and proteostasis. Such a state is consistent with broad shutdown of core metabolic pathways (carbon metabolism, glycolysis/gluconeogenesis, fatty acid metabolism and related catabolic processes) and with the induction of integrated stress response signaling. Importantly, this late metabolic collapse is not merely an in vitro curiosity but resonates with clinical observations. Serum metabolomic profiling of hospitalized H7N9 patients has suggested inhibition of fatty acid

metabolism during infection, with specific fatty-acid-related metabolites (including palmitic acid, erucic acid and phytal) negatively correlating with the extent of lung inflammation and disease severity, and with disrupted expression of lung epithelial integrity genes in biopsy material⁴.

The concordance between our epithelial transcriptomic suppression of metabolic programs and patient level associations supports the idea that H7N9 severity may be linked to a failure to maintain metabolic and epithelial barrier homeostasis during the course of infection. Notably, the patient metabolomics study also emphasizes epithelial integrity and lung inflammation as central pathological features, consistent with the view that epithelial collapse can amplify downstream immunopathology rather than simply reflecting a passive consequence of viral replication⁴.

Our module results also highlight a potential mechanistic bridge between stress responses and antigen presentation. Stress induced translational control and trafficking perturbations can impact the synthesis, glycosylation, and surface delivery of antigen presentation machinery. Indeed, influenza viruses have been shown to downregulate MHC I during late infection through distinct mechanisms: IAV can drive a global loss or degradation of MHC I within infected cells, whereas other influenza lineages can reduce surface expression by intracellular retention and delayed trafficking⁵.

Although this literature focuses primarily on MHC I and later time points, it supports the broader principle that influenza infection can reshape antigen presentation capacity by altering transcription, trafficking, and protein stability. In our dataset, the combined reduction of ER/Glyco/Lys and MHC II module scores in H7N9 at 24 hpi is consistent with a model in which late disruption of secretory/endomembrane homeostasis contributes to reduced antigen presentation potential. The early suppression of MHC II-associated GO terms suggests that H7N9 may additionally engage earlier transcriptional repression or regulatory interference upstream of trafficking collapse, producing a two-phase phenotype: an early “immune communication deficit” followed by a late “bioenergetic/secretory deficit”.

An additional layer of H7N9 specificity is captured by our supplementary analyses in which we excluded mock controls and instead focused on genes commonly regulated across viruses but quantitatively more extreme in H7N9. Those analyses indicate that H7N9 specificity is partly quantitative: many pathways are engaged by other viruses but are more strongly shifted in H7N9 (greater induction of transcription/stress programs and stronger suppression of immune-associated modules). This “setpoint shift” fits the idea that severe H7N9 reflects dysregulated coordination with a strong antiviral signaling and inflammation, but suboptimal immune communication and homeostatic maintenance.

The enrichment of pathways annotated to other viral infections (e.g., HSV-1) likely captures shared host-response backbones rather than HSV-specific biology, notably, HSV-1 and influenza A induce overlapping RNA-processing and nuclear stress programs such as NEAT1_2 (paraspeckle) remodeling and circRNA-related signatures⁶. In our dataset, this supports the idea that a broad antiviral transcriptional program can still coexist with reduced antigen-presentation module activity and declining homeostatic capacity in H7N9.

Early DEG calls are power-limited and enrichment terms can reflect annotation overlap in an epithelial system. Calu-3 captures epithelial-intrinsic states but not multicellular immune feedback in vivo.

In conclusion, time-resolved cross-strain profiling indicates that H7N9 is distinguished by early suppression of antigen-presentation programs and late ISR-linked metabolic and secretory collapse, nominating stress–metabolism–presentation nodes as candidate drivers of severe disease.

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Discussion – Fangyi CHEN

Avian influenza A(H7N9) remains a clinically important zoonotic threat because severe human lower respiratory tract disease can occur even when epidemic strains are low pathogenic in poultry, highlighting that pathogenic potential is shaped not only by viral replication but also by how host responses are orchestrated across time and immune axes.^[1] In this study, we leveraged a time-resolved, cross-strain Calu-3 epithelial infection framework to separate conserved influenza response components from strain-specific programs, with a specific focus on the temporal organization of H7N9-associated dysregulation.

Time-structured model: early innate activation with suppressed antigen presentation

A key implication of our time-resolved H7N9 analysis is that epithelial antiviral activation and antigen-presentation machinery do not necessarily rise in parallel. Across early and intermediate time points (3–12 h), enrichment of downregulated genes repeatedly highlighted MHC class II complex assembly and antigen processing and presentation via MHC class II, indicating that antigen-presentation programs are transcriptionally suppressed from the earliest stages and remain repressed through mid-infection. In the same temporal window, upregulated programs were dominated by defense response to virus and type I interferon signaling, consistent with robust epithelial innate activation.

Although airway epithelial cells are not professional antigen-presenting cells, multiple studies support inducible epithelial MHC-II expression under inflammatory conditions and a role for barrier epithelia in shaping local immune programming.^[2] From this perspective, early and sustained suppression of MHC-II-associated transcription in epithelium could influence mucosal immune coordination, including the timing and quality of local CD4+ T cell priming and epithelial-immune crosstalk. This interpretation is aligned with patient studies reporting reduced antigen-presenting capacity in myeloid compartments during severe H7N9, such as decreased monocyte HLA-DR associated with disease severity.^[3] While our dataset does not include immune cells and therefore cannot directly link epithelial MHC-II transcriptional suppression to systemic antigen-presentation defects, the concordant patterns across compartments motivate a model in which H7N9 pathogenesis involves attenuation of antigen presentation in both epithelial and myeloid contexts, occurring alongside heightened innate inflammatory outputs.

Late-stage stress-metabolic remodeling and ER-lysosome dysfunction

A second central theme is a marked late-stage transition at 24 h, when downregulated genes become overwhelmingly enriched for broad metabolic processes (including fatty acid, amino acid, and organic acid metabolism), while upregulated genes emphasize cellular stress pathways including integrated stress response (ISR) signaling. This transition is also supported by progression-based contrasts (24 h vs 3 h), which show sustained induction of canonical interferon-stimulated and inflammatory mediators and concurrent repression of core metabolic enzymes, consistent with a trajectory in which inflammatory outputs persist while metabolic homeostasis deteriorates.

These observations fit within a broader literature describing immune-metabolic remodeling during influenza infection, in which host defense, viral replication, and tissue integrity compete for central carbon metabolism, lipid handling, and mitochondrial capacity.^[4] Notably, our module-level framework further suggests that H7N9 uniquely couples coordinated decreases in Metabolism and ER_Glyco_Lys modules with reduced MHC_II activity and strong ISR induction at 24 h. Given known interfaces among lysosomal function, autophagy flux, antigen processing, and inflammatory regulation, ER-glyco-lysosome disruption

represents a plausible mechanistic bridge linking metabolic stress to impaired processing capacity and maladaptive inflammation. ISR activation itself is also positioned as a convergence node in virus infection biology, with eIF2 α phosphorylation reshaping translation, amino acid metabolism, and cytokine outputs.^[5] Importantly, transcriptomic inference alone cannot determine whether ISR is protective or pathogenic in this context, making functional directionality a priority for follow-up experiments.

Reconciling inflammatory mediator signatures with strain-specific pathway organization

Clinical studies of H7N9 have repeatedly noted elevated inflammatory mediators including IL-6 and CXCL10 (IP-10), often associated with severity, supporting a hyperinflammatory component.^[6] Our temporal progression analysis likewise detects increasing inflammatory mediator expression toward later time points. Yet, within the strain-specific functional landscape at 24 h, H7N9 is distinguished not only by classical interferon/cytokine pathway dominance but also by transcriptional regulation, cell-cycle-linked programs, and lysosome-associated terms among upregulated strain-specific genes, paired with downregulated metabolic and antigen-presentation modules. Together, these findings support a model in which H7N9-associated inflammatory outputs may be amplified or prolonged in part because homeostatic buffering capacity is compromised, rather than solely because cytokine induction is stronger in a simple quantitative sense.

Inflammasome biology provides an additional axis to integrate. In vivo work has implicated NLRP3–caspase-1 signaling in H7N9 immunopathology, and genetic or pharmacologic disruption of IL-1–linked inflammasome outputs can attenuate disease severity in mouse models.^[7] Although epithelial transcriptomes cannot directly report inflammasome activation states that are often post-transcriptional and cell-type specific, the late stress and metabolic disruption we infer could plausibly contribute to inflammasome priming or impaired resolution in infected lung tissue, particularly within myeloid populations. Testing this cross-compartment coupling requires integrated models.

Limitations and interpretive boundaries

Several limitations frame interpretation and motivate the next experimental stage. First, Calu-3 cells provide a controlled human airway epithelial model for cross-strain comparisons, but they do not recapitulate airway cellular diversity or include professional antigen-presenting cells and recruited immune compartments that are central to antigen presentation and inflammatory amplification in vivo. Second, our conclusions derive from transcriptomics; antigen presentation, ISR signaling, lysosomal flux, and metabolic capacity are heavily regulated at protein, organelle, and metabolite levels, so mechanistic claims must be validated with orthogonal functional assays. Third, the time window ends at 24 h. The dominance of 24 h–specific DEGs in H7N9 and the limited persistence of a small core response imply that later kinetics (48–72 h) may be essential for distinguishing delayed emergence from late collapse and for linking host state transitions to viral burden and cytopathology. Finally, differential expression detection depends on statistical power. For example, the near absence of VN1203/H5N1 DEGs at 7 h under a gene-wise t-test approach is consistent with reduced sensitivity in small-n settings without empirical Bayes variance moderation, and thus should be interpreted cautiously rather than as definitive biological quiescence.

Future directions: concrete experimental designs

Our results motivate a focused roadmap to move from descriptive dynamics to causality,

prioritizing three mechanistic axes: antigen presentation, immunometabolism/ER–lysosome function, and the integrated stress response (ISR). We will first validate epithelial antigen-presentation suppression by quantifying CIITA and core MHC-II components across the time course with IFN- γ inducibility controls, then link these measurements to functional antigen-presentation readouts and, where feasible, viral-factor perturbations.^[8] We will next determine whether late metabolic and ER–glyco–lysosome signatures correspond to measurable functional deficits using integrated metabolic profiling plus targeted lysosome/autophagy assays, and test causality with rescue/perturbation designs.^[9] Finally, we will establish ISR directionality by quantifying canonical ISR markers and perturbing ISR signaling while mapping contributions from the four major eIF2 α kinases to clarify upstream triggers and therapeutic implications.^[10] Key observations will be validated in physiologic airway systems, including primary air–liquid interface cultures and epithelial–immune co-cultures, to test cross-compartment coupling not captured by epithelial-only models.

In parallel, we will pursue translational extensions that explicitly leverage temporal staging. First, we will evaluate a late-stage composite signature reflecting stress–metabolic remodeling together with antigen-presentation suppression as a candidate biomarker of severe trajectories by testing reproducibility and severity associations in independent patient datasets. Second, we will interrogate baseline-state effects, including whether epithelial differentiation status or pre-existing metabolic state shifts the timing and magnitude of the 12→24 h transition, which could explain inter-individual heterogeneity and enable stratified hypotheses for validation. Third, our model motivates therapeutic timing as a primary variable: early interventions may prioritize antiviral reinforcement and containment of initial innate activation, whereas later interventions may be more effective if aimed at stabilizing stress responses, metabolic capacity, and ER–lysosomal processing. These directions provide a stage-aware framework for host-directed strategies while remaining testable with the experimental systems described above.

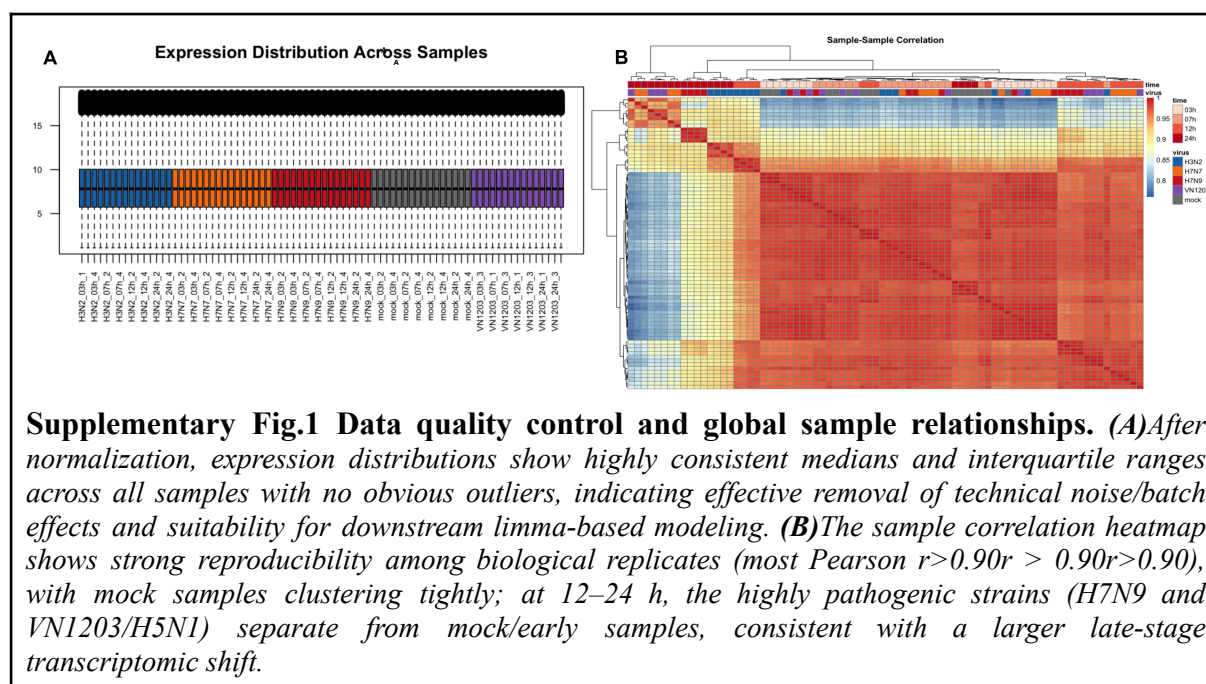
Beyond H7N9, this work illustrates why time-resolved and cross-strain comparative designs are necessary for mechanistic inference in respiratory virus immunology. Analyses limited to single time points can obscure whether a pathway is absent or delayed and can conflate early antiviral defense with late stress-associated collapse. Our data support a framework in which H7N9 pathogenesis is better conceptualized as a sequence of staged host states that increasingly decouple innate inflammatory outputs from antigen-presentation capacity and metabolic homeostasis. This “decoupling” model is valuable because it is experimentally actionable: it predicts that interventions restoring antigen processing or stabilizing metabolic and ER–lysosomal function could reshape inflammatory trajectories even without directly targeting canonical interferon pathways.

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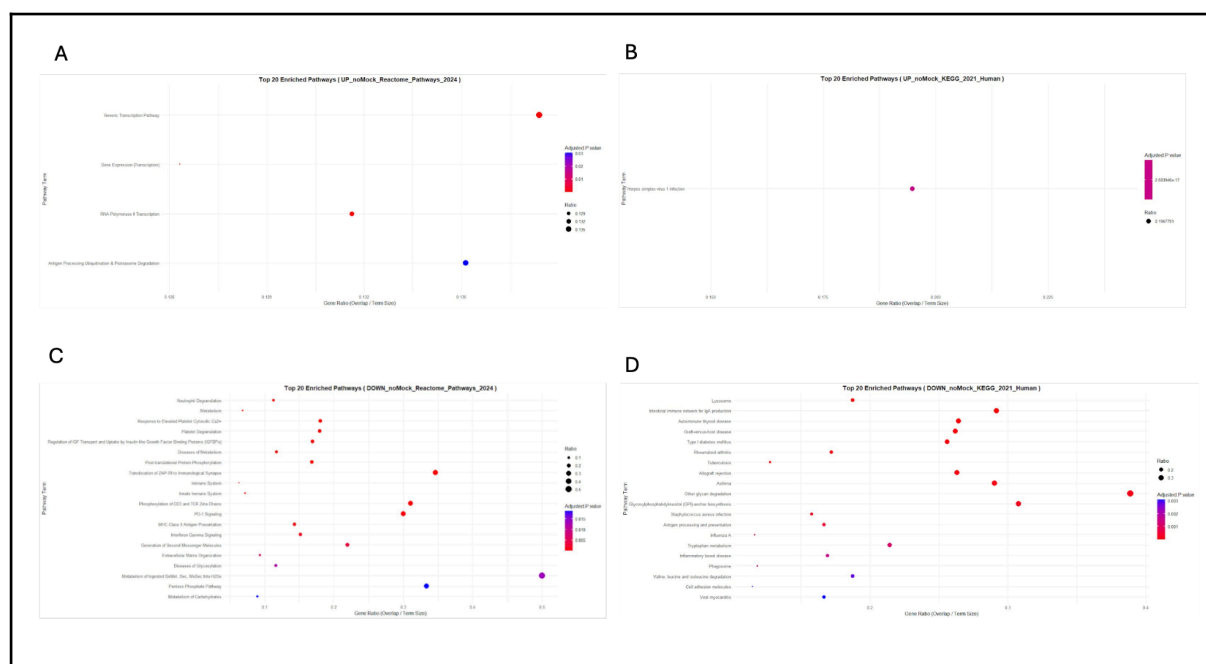
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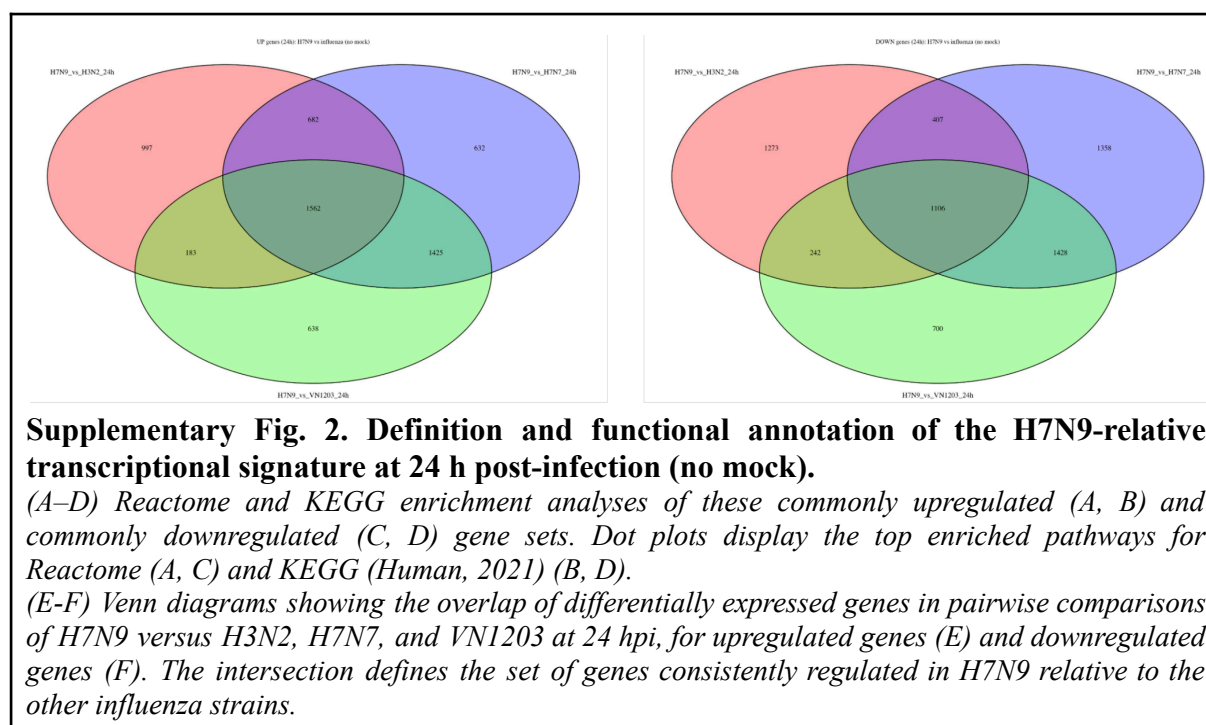
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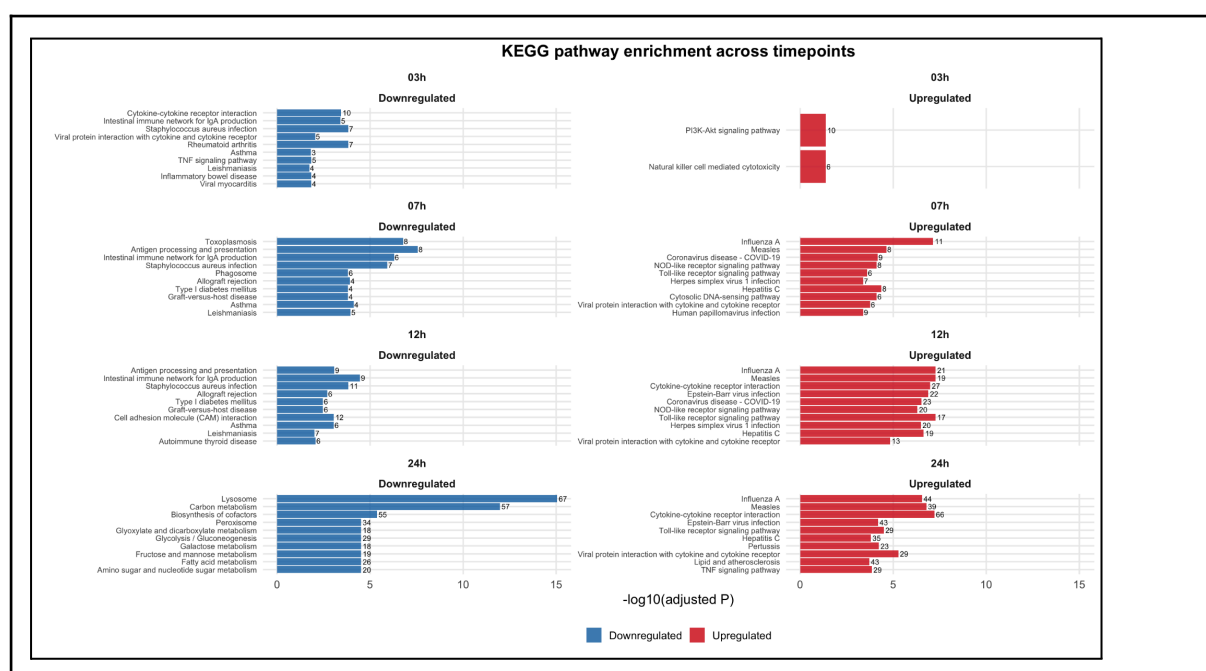
Supplementary Fig.1 Data quality control and global sample relationships. (A)After normalization, expression distributions show highly consistent medians and interquartile ranges across all samples with no obvious outliers, indicating effective removal of technical noise/batch effects and suitability for downstream limma-based modeling. (B)The sample correlation heatmap shows strong reproducibility among biological replicates (most Pearson $r > 0.90$), with mock samples clustering tightly; at 12–24 h, the highly pathogenic strains (H7N9 and VN1203/H5N1) separate from mock/early samples, consistent with a larger late-stage transcriptomic shift.

Following normalization, all 40 libraries exhibited highly comparable global expression distributions with no prominent outliers, supporting the distributional assumptions required for robust differential expression analysis using the limma framework. Sample–sample correlation and hierarchical clustering further demonstrated high consistency among biological replicates and a stable baseline defined by mock-infected samples across time points. While early responses (03–07 h) were broadly similar across influenza strains and closer to mock controls, the highly pathogenic viruses (H7N9 and VN1203/H5N1) showed clear divergence at 12–24 h, indicating pronounced and strain-specific remodeling of host transcriptomic signatures over time.





Enrichment of the commonly upregulated genes revealed a functional profile dominated by **transcriptional activity, host stress responses and intracellular signaling pathways**, indicating that H7N9 infection is associated with enhanced engagement of transcriptional programs relative to other influenza viruses at 24 hours post-infection. In contrast, the commonly downregulated gene set was enriched for **immune-associated pathways**, including innate immune effector functions and signaling pathways involved in lymphocyte activation. Notably, these pathways were not absent in the responses to other viruses but were **relatively attenuated in H7N9-infected cells**, consistent with a quantitative dampening of immune processes rather than their complete suppression.



Supplementary Figure S3. KEGG pathway enrichment across time points during H7N9 infection. KEGG pathway enrichment analysis of differentially expressed genes (DEGs) in H7N9-infected Calu-3 cells relative to time-matched mock controls at 03 h, 07 h, 12 h, and 24 h post-infection. For each time point, enriched pathways are shown separately for downregulated (blue, left panels) and upregulated (red, right panels) gene sets. Bars represent the significance of enrichment expressed as $-\log_{10}(\text{adjusted } P \text{ value})$, calculated using Benjamini–Hochberg correction. Numbers displayed at the end of each bar indicate the number of DEGs contributing to the corresponding pathway. Only pathways with adjusted $P < 0.05$ are shown.

Early time points are characterized by suppression of antigen presentation–related pathways alongside activation of innate immune and antiviral signaling. By 24 h post-infection, downregulated pathways are dominated by metabolic and lysosomal processes, whereas upregulated pathways remain enriched for inflammatory and cytokine-mediated signaling.

Module	Genes (symbols)
ER Glyco Lys	HSPA5, PDIA3, CANX, CALR, SEC61A1, SSR1, STT3A, STT3B, LAMP1, LAMP2, CTSB, CTSD, CTSL, CTSS, ATP6V1A, ATP6V0D1
ISR	ATF4, ATF3, DDIT3, PPP1R15A, ASNS, TRIB3, CHAC1, GADD45A
Metabolism	HK1, HK2, PFKP, PFKM, GPI, ALDOA, GAPDH, PGK1, ENO1, PKM, LDHA, G6PD, PGD, TKT, TALDO1, PDHA1, CS, IDH3A, SDHA, FH, MDH2, ACACA, FASN, SCD, ELOVL6, CAD, DHODH, PPAT, ATIC, IMPDH1, IMPDH2
MHC I	HLA-A, HLA-B, HLA-C, B2M, TAP1, TAP2, TAPBP, ERAP1, ERAP2, PSMB8, PSMB9, NLRC5
MHC II	CIITA, HLA-DRA, HLA-DRB1, HLA-DPA1, HLA-DPB1, HLA-DQA1, HLA-DQB1, CD74, CTSB, CTSD, CTSL, CTSS

Table 1:| Gene modules used for transcriptomic module scoring.

The table lists the gene symbols included in each predefined module (*ER_Glyco_Lys*, *ISR*, *Metabolism*, *MHC_I*, and *MHC_II*) that were used for module-based analyses. Each module consists of the indicated genes and was applied consistently across all samples and time points.